INCH-POUND
MIL-DTL-24643/19E
22 August 2002
SUPERSEDING
MIL-C-24643/19D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS6SGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, (see table I for size).
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5 or standard identification code method 1 (two conductors each A, B and C).
Third - The six conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly. Cabling sequence shall be A, B and C.
Fourth - A binder or combination binder and barrier.
Fifth - Cross-linked polyolefin jacket. (see table I for thickness)

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TABLE I. Details.

Military part no. M24643/19	Size	Conductor size AWG	Type's overall diameter (inches) LS6SGU min	Type's overall diameter (inches) LS6SGU max	Diameter over outer insulation tape nominal (in.)	Conductor resistance per 1000 feet at 25 deg C max (ohms)	Jacket thickness min avg (in.)	Insulation resistance (megohms)	Cold bending mandrel dia max (in.)	Accelerated service loading (ampere)
-01UN	100	0 (class D)		1.600	0.453	0.106	0.060	160	20	475
-02UN	125	00 (class D)		1.790	0.507	0.0850	0.060	150	22	520
-03UN	150	000 (class D)		1.960	0.557	0.0674	0.060	135	24	570
-04UN	200	0000 (class D)		2.200	0.634	0.053	0.075	125	27	650

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EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	5000
Insulation resistance - megohms- 1000 feet, minimum	
Conductor to conductor	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Cold working (minus 20 ± 2°C	No damage
Gas flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 800 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-017)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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